Yun-Rou (Lilian), Lin

+886 988929049| tao86419@gmail.com
Yunrou.github.io | linkedIn/Yunrou | github/Yunrou

EDUCATION

Master of Computer Science 2019 - 2022
NATIONAL YANG MING CHIAO TUNG UNIVERSITY
GPA: 4.28/4.3
The Phi Tau Phi Scholastic Honor Society of the Republic of China Honorary Membership

Bachelor of Computer Science 2015 - 2019
NATIONAL CHIAO TUNG UNIVERSITY
GPA: 3.9/4.3
Teaching Assistant (TA): Numerical Methods, 2019
Vice Mentor @NCTU Blue Gear Program, Minister of Artistic Design Dept. @NCTU CS Student Associate, Director @NCTU UU Club

EXPERIENCE

SYNOPSYS INC - APPLICATION ENGINEERING, STAFF ENGINEER JAN 2025 – PRESENT
Delivered **ICV dummy training** from runset structure, layer operations to unified fill.
Prototyped a Runset Generator for **simple non-tabu DRC rules** from DRM with the use of **Lark parser and transformer**.
Integrated **timing-aware fill** flow from **RC extraction**, **STA** to **Fill** in both Python and C/C++ version.
Prototyped an **HGNN model** that formulates design into a graph and optimizes it through chunking.
Delivered **GH Copilot** set up training to 60+ signoff team members.
FY26 PurplePoster - Standalone ICV Timing-Aware Fill
FY25 PurplePoster - Fillable Region Smooth for Signoff Fill Runtime and Quality

SYNOPSYS INC - APPLICATION ENGINEERING, SR. ENGINEER AUG 2022 - JAN 2025
ICV runset coding and qualification for dummy fill insertion in advanced process.
Optimized flow that achieves **10X runset coding effort improvement**.
Developed utilities that reduce **30% runtime** and improve QoR in FEOL dummy.
Identified and debugged design rule violations before tapeout, performed RCA, and implemented effective solutions.
Collaborated with the foundry and RD to define requirements and validate cutting-edge features
FY24 PurplePoster - Accelerate TSMC N2 runset development flow for expandable fill

PUBLICATION

NAVIGATING COLOR CONSTRAINTS IN MULTI-VIEW VISUALIZATIONS WITH MVCOLOR 2025
IEEE PACIFICVIS

RESEARCH

MVCOLOR: RECOMMENDATION OF COLOR ENCODINGS FOR MULTI-VIEW VISUALIZATIONS
GRAPHICS AND PERCEPTION LAB
Sep 2019 – Apr 2022
Developed an interactive recommendation system that models the colorization of a multi-view visualization system.
Formulated a grouping problem into **Set Partitioning Problem** solved by **Gurobi** optimization.
Ensured color discriminability using a **Genetic Algorithm**.
Used **React** for front-end and **Django/Django REST Framework** for back-end.

LANGUAGES

Chinese: Native
Japanese: A2
English: B2 TOEFL iBT: 89 (R24/L23/S21/W21)

SKILLS

Programming Languages: C/C++, Python, Perl, R, MATLAB
Web Development: HTML/CSS/JS, React, Django/Django REST Framework
Visualization: D3.js, Matplotlib, Seaborn
Technology: Git/GitHub, LaTeX, Cursor, cron-job